

Day 35 - Machine Learning

1) What is Machine Learning?

Machine Learning is a branch of Artificial Intelligence in which machines learn from data and make predictions or decisions without being explicitly programmed.

Example: Spam detection, recommendation system, house price prediction.

2) Difference between AI and ML

Artificial Intelligence (AI):

AI is a broad concept that makes machines think and act like humans.

Machine Learning (ML):

ML is a part of AI where machines learn from data.

Difference:

- AI is the bigger field.
- ML is a subset of AI.
- AI includes reasoning and decision-making.
- ML mainly focuses on learning patterns from data.

In short:

AI is the parent, ML is the child.

3) What are the different types available in ML?

There are mainly 3 types of Machine Learning:

1. Supervised Learning
2. Unsupervised Learning
3. Reinforcement Learning

4) What is the life cycle of ML?

The Machine Learning life cycle is the complete process of building an ML model.

Steps in ML Life cycle:

1. Problem Definition
2. Data collection
3. Data Preprocessing
4. Exploratory Data Analysis
5. Feature Engineering
6. Model Selection
7. Model Training
8. Model Evaluation
9. Deployment
10. Monitoring

In short:

collect data → clean data → Train model →

Test model → Deploy

5) What is Structured Data?

Structured data is the data which is arranged in rows and columns in a fixed format.

Examples:

- Excel sheet
- CSV file
- SQL database

Example: Name, Age, Salary table.

6) What is Unstructured Data?

Unstructured data is the data which does not have a fixed format.

Examples:

- Images
- Videos
- Audio
- Emails
- Text documents

7) What are the applications of ML?

Machine Learning is used in many real-world applications such as:

- Recommendation systems
- Spam email detection
- Face recognition
- Fraud detection
- Medical diagnosis
- Chatbots
- Self-driving cars
- Stock prediction
- Speech recognition

8) What is Supervised Learning? Explain about the types

Supervised Learning is a type of machine learning in which the model is trained using labeled data.

That means the input and correct output are already known.

Types of Supervised Learning:

1. Classification

Classification is used when the output is in categories or classes.

Examples:

- Spam / Not Spam
- Pass / Fail

- Yes / No

2. Regression

Regression is used when the output is a continuous numerical value.

Examples:

- House price prediction
- Salary prediction
- Temperature prediction

In short:

Supervised Learning learns from labeled data and has two types: Classification and Regression.

9) What is Unsupervised Learning? Explain about the types

Unsupervised Learning is a type of machine learning in which the model is trained using unlabeled data.

Here, the system finds patterns on its own.

Types of Unsupervised Learning:

1. Clustering

Clustering means grouping similar data points together.

Example:

customer segmentation

2. Association

Association means finding relationships between items.

Example:

If a customer buys bread, they may also buy butter.

3. Dimensionality Reduction

It means reducing the number of features while keeping important information.

Example:

Reducing 100 columns into 10 useful columns.

In short:

Unsupervised learning works on unlabeled data and includes clustering, association, and dimensionality reduction.

10) What is Feature Engineering?

Feature Engineering is the process of selecting, creating, or transforming useful features from raw data to improve model performance.

Examples:

- Converting date into day, month, year
- Creating age group from age

- Creating BMI from height and weight

Importance:

Good features improve model accuracy.

11) What is Model Evaluation?

Model Evaluation is the process of checking how well a machine learning model performs.

It helps us know whether the model is accurate and reliable.

Common evaluation metrics:

- Accuracy
- Precision
- Recall
- F1-score
- MAE
- RMSE

12) What is Preprocessing?

Preprocessing is the process of cleaning and preparing raw data before giving it to the machine learning model.

It includes:

- Handling missing values
- Removing duplicates

- Encoding categorical data
- Feature scaling
- Removing outliers

Importance:

Clean data gives better results.

13) What is Confusion Matrix?

Confusion Matrix is a table used to evaluate the performance of a classification model.

It compares the actual values with predicted values.

It contains 4 parts:

1. True Positive (TP) - correctly predicted positive
2. True Negative (TN) - correctly predicted negative
3. False Positive (FP) - Incorrectly predicted positive
4. False Negative (FN) - Incorrectly predicted negative

Use:

It helps measure classification accuracy.

14) What is correlation?

correlation shows the relationship between two variables.

It tells whether one variable changes when another variable changes.

Types of correlation:

1. Positive correlation - Both increase together
2. Negative correlation - One increases, other decreases
3. Zero correlation - No relationship

correlation range:

- $+1$ = Perfect positive
- 0 = No relation
- -1 = Perfect negative

15) Explain the Evolution of ML

The evolution of ML means how machine learning developed over time.

Stages:

1. Rule-based systems
2. Statistical learning
3. Traditional ML algorithms
4. Big data era
5. Deep learning era

6. Modern AI and Generative AI

In short:

ML has evolved from simple rule-based systems to advanced intelligent systems.

16) What is the History of ML?

Machine Learning has developed over many years.

Important history:

- 1950 - Alan Turing introduced the idea of intelligent machines
- 1952 - Arthur Samuel created a self-learning checkers program
- 1957 - Perceptron model was introduced
- 1980s - Neural networks became popular
- 1990s - Statistical learning methods grew
- 2010s - Deep Learning became successful
- 2020s - Generative AI became popular

17) What is Model Selection?

Model Selection is the process of choosing the best machine learning algorithm for a problem.

Examples of models:

- Linear Regression

- Decision Tree
- Random Forest
- SVM

Factors for model selection:

- Accuracy
- Speed
- complexity
- Type of data

18) How to train the data?

Training the data means teaching the model using the training dataset.

Steps:

1. collect the dataset
2. Preprocess the data
3. Split into training and testing data
4. select an algorithm
5. Train the model using training data

Python Example:

```
model.fit(x_train, y_train)
```

This helps the model learn patterns from the data.

19) How to test the data?

Testing the data means checking the performance of the model on unseen data.

Steps:

1. Use test data
2. Predict output
3. Compare predicted output with actual output

Python Example:

```
y_pred = model.predict(x_test)
```

Testing helps us know how well the model works.

20) What is cross validation?

Cross Validation is a technique used to evaluate a machine learning model more accurately.

Most common type: K-Fold Cross Validation

- Data is divided into K parts
- Model is trained on K-1 parts
- Tested on 1 part
- This process repeats K times

Importance:

- Gives better evaluation
- Reduces bias

- Helps avoid overfitting

21) Why Regularization Techniques are Important?

Regularization techniques are important because they help prevent overfitting.

Overfitting means:

The model learns training data too much and performs poorly on new data.

Regularization helps by:

- Reducing complexity
- Improving generalization
- Making model stable

Types:

- L1 Regularization (Lasso)
- L2 Regularization (Ridge)
- Elastic Net

22) How do we Handle Missing Values?

Missing values are empty or null values in a dataset.

Ways to handle missing values:

1. Remove missing data

Delete rows or columns with missing values.

2. Fill with Mean

Used for numerical data.

3. Fill with Median

Useful when there are outliers.

4. Fill with Mode

Used for categorical data.

5. Forward Fill / Backward Fill

Used in time-series data.

6. ML-based Imputation

Advanced method for filling missing values.